

Wen-Chieh Chao, MD MS

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Ultrasound-responsive biomaterials • 3D bioprinting

EDUCATION

Stanford University

2024 - 2026

Masters of Science - BioEngineering (GPA: 3.93)

Stanford, CA

- Computational Coursework:
Machine Learning, Cloud Storage and Computing, Signal Processing, Feedback Control Design
- Biomedical Coursework:
Quantitative Physiology, Systems Biology, Computational Structural Biology, Tissue Engineering, Biomaterials
- Entrepreneurship Coursework:
Patent Law and Strategy, Biodesign and Innovation

National Taiwan University

2017 – 2023

Doctor of Medicine - MD, Medicine (GPA: 4.02)

Taipei, Taiwan

- Clinically trained with full two-year hospital rotations across 30+ departments/specialties
- Provided primary care for 80+ hospitalized patients; assisted in 20+ and observed 150+ surgeries.

RESEARCH EXPERIENCE

Graduate Researcher

Oct 2024 – Present

The Skylar-Scott Lab (PI: Mark Skylar-Scott), Stanford - Biomaterials, Liposomes

Stanford, CA

- Enabled controllable, ultrasound-triggered fibrin crosslinking by engineering a thrombin-loaded, ultrasound-responsive liposome release platform.
- Achieved robust and tunable thrombin release by optimizing membrane mechanics across >100 liposome fabrication conditions.
- Performed iPSC culture experiments to evaluate the effects of a lab-developed engineered growth factor.

Research Assistant

Dec 2023 - Jun 2024

Wu Lab (PI: Yu-Wei Wu), IMB Academia Sinica - Neuroanalysis, Neuro-behavior

Taipei, Taiwan

- Characterized DBS-induced behavioral and neural changes in Parkinson's mouse models by conducting controlled behavioral assays on >20 PD mice.
- Increased reproducibility and throughput of dendritic spine quantification by optimizing DeepD3-based analysis workflows.
- Extracted nonlinear structure in neural datasets by using PCA/UMAP-based dimensionality-reduction analyses.

Research Assistant

Apr 2021 - Jun 2021

Brain and Mind Lab(PI: Joshua O. Goh), National Taiwan University - fMRI, VR design

Taipei, Taiwan

- Identified age-related differences in brain activation by performing full fMRI preprocessing and 2nd-level SPM analysis in MATLAB.
- Enabled controlled testing of spatial navigation behavior by building non-Euclidean virtual environments in Unity.
- Reduced experimental analysis time by 80% by integrating open-source gaze-analytics pipelines into the workflow.

Rotation Internship

Oct 2022 - Nov 2022

The Tytgat Institute(PI: Stan van de Graaf), Amsterdam UMC - Receptor screening

Amsterdam, Netherlands

- Screened bile-salt-responsive GPCR candidates by applying the Presto-Tango reporter platform in HTLA cells.
- Boosted transfection success rate from 50% to 90% by optimizing reagent stoichiometry and workflow conditions.
- Supported project planning by synthesizing bile-salt signaling literature & presenting updates to research team.

PUBLICATIONS & PRESENTATIONS

Betty Cai, Seunghoon Lee, **Wen-Chieh Chao**, Mark Skylar-Scott, Sarah C. Heilshorn *Ultrasound-Sensitive Liposomes for Triggered Thrombin Release*. Poster presentation within program, April 16, 2025.

Wang, L. S., Lee, Y. H., Chuang, J. Y., Wang, T. S., Wang, P. K., **Chao, W. C.**, Chen, C. Y., Su, Y. S., Goh, J. O. S. (2023). *Greater distance judgment distortion is associated with more subjective than objective representational similarity in extrastriate functional responses in older compared to younger adults*. PST570.30/VV54. Poster presentation at the Annual Meeting for the Society for Neuroscience, Washington, D.C., USA. Abstracts: [URL]; Poster: [URL].

Wang, P. K., Lee, Y. H., Chuang, J. Y., Wang, C. L. S., **Chao, W. C.**, Chen, C. Y., Su, Y. S., Goh, J. O. S. (2023). *Different strategic neural correlates representing distances implicates age-related distortions in spatial navigation*. Poster presentation at the Annual Meeting for the Cognitive Neuroscience Society, San Francisco, USA. [URL].

TECHNICAL SKILLS

Wet Lab & Biomaterials: Liposome fabrication, purification, and controlled cargo release; iPSC culture

Computational & Modeling: MATLAB, Python, fMRI/SPM, neural signal analysis, PCA, UMAP

Instrumentation & Imaging: Ultrasound, confocal/fluorescence microscopy, FIJI, DeepD3